

COURSE INTRODUCTION

DATA WRANGLING AND VISUALIZATION 2026/27

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WHY DATA WRANGLING AND VISUALIZATION?

University courses cover a wide range of **data analysis**, **statistics**, and **machine learning methods**

However, practical data science requires essential skills that occur before and after analysis:

1. **Data Wrangling** (before analysis)

- Cleaning, reshaping, and structuring raw, messy data
- Preparing tidy datasets ready for modeling

2. **Data Visualization** (during and after analysis)

- Exploring patterns and validating analytical assumptions
- Designing clear visual representations of data

3. **Data Communication** (final output)

- Integrating charts and insights into reports, articles, and presentations
- Communicating results effectively to any audience

WHAT YOU WILL LEARN

The course teaches how to

- clean, structure, and transform raw data into tidy data
- produce meaningful visual representations of the data
- communicate data-driven insights effectively

These are foundational skills required for anyone working with data or pursuing a career in data science

The course emphasizes a hands-on approach to build a **reproducible data pipeline** from scratch:

- Real-world practice working directly with actual datasets
- A complete workflow connecting raw data directly to final reports
- Fully reproducible, automated, and reusable code

CONTENTS

1. Data Wrangling

- Structuring and reshaping raw data into tidy formats
- Data transformation, filtering, and joining
- Assessing and resolving data quality issues

2. Data Visualization

- Core principles of effective visual design
- Building charts with the grammar of graphics
- Visual integrity and avoiding misleading graphics

3. Data Communication

- Communicating numbers and statistical findings
- Presenting risk, probability, and uncertainty effectively
- Best practices from data journalism and public communication

COURSE SEQUENCE

Although the natural data pipeline starts with wrangling, the course begins with **data visualization** for several key pedagogical reasons:

- Immediate engagement through visual feedback from day one
- Clear motivation for wrangling, showing why and how raw data must be reshaped
- Time for visual design principles to mature before moving to data communication

The course sequence then moves to **data wrangling**, revisits visualization with structured datasets, and concludes with **data communication**

ORGANIZATION

The course consists of 24 lectures scheduled as follows:

Day	Time	Room
Tuesday	15:45 – 17:15	Computer Labs 3 + 5
Friday	14:00 – 15:30	Room 2

The course combines

- traditional lectures
- computer labs
- case study discussions

Computer practicals will be conducted using **R** and **RStudio**

You are strongly encouraged to **bring your laptop** and **actively participate** in classes and labs

MOODLE MATERIALS

Materials that will be made available through Moodle:

- Slides
- Complementary readings
- R labs
- Case studies

There will be a **course diary** with a week-by-week summary of the classes

You will receive email notifications via Moodle announcements:

- whenever new materials are uploaded
- whenever published materials are updated

Materials will be published before the corresponding classes

OFFICE HOURS AND CONTACTS

Office Hours

- When: Tuesday 12:30 – 14:30
- Where: Office Z.A11 (1st floor, Building Zeta A)

Variations will be announced on Moodle and my university webpage

Email to cristiano.varin@unive.it:

- Email should be used exclusively for **organizational matters**
- Course content questions will be addressed during office hours or after class, **not via email**

Discussing course materials and labs via email often leads to misunderstandings best resolved in person

ASSESSMENT

Evaluation consists of a two-step exam:

1. Practical computer exam
2. Oral exam

The oral exam is reserved for students who pass the practical exam

The practical computer exam requires implementing a complete data wrangling, visualization, and communication pipeline on a real dataset

The oral exam evaluates your level of mastery of the course topics

Sample exams will be provided and discussed in class, alongside a mock practical exam simulation